

Network synthesis

Types of N/w

- i) LC Network :- $Z(s)$ or $Y(s)$
- ii) RC Network :- $Z(s)$ or $Y(s)$
- iii) RL Network :- $Z(s)$ or $Y(s)$

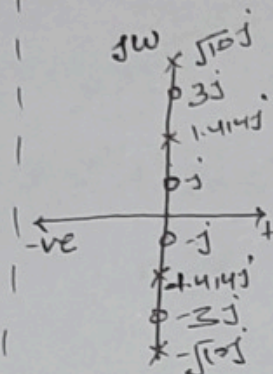
LC Network

* Having only L & C $\rightarrow$ R X

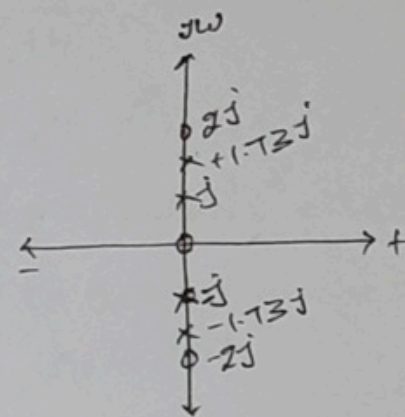
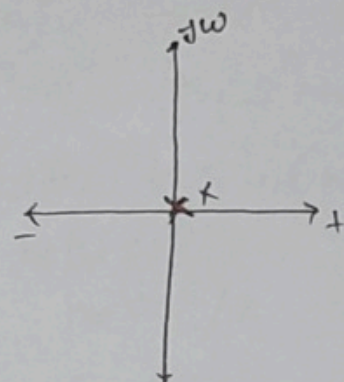
Properties of LC N/w $T(s) = \frac{N(s)}{D(s)}$

- i) Poles & zero are having imaginary value.
- ii) Poles & zero are lying on $j\omega$ axis in alternate fashion.
- iii) Highest power of $N(s)$ & $D(s)$ differ by unity (most)
- iv) lowest power of $N(s)$ & $D(s)$ must differ by unity.

Pole-zero plot of example



X



Example

i) $F(s) = k \frac{(s^2+1)(s^2+9)}{(s^2+2)(s^2+10)}$ $Z = \pm j, \pm 3j$
 $P = \pm \sqrt{2}j, \pm \sqrt{10}j$

ii) $F(s) = \frac{s^5 + 3s^3 + 9s}{7s^4 + 6s^2}$ $= \frac{s(s^4 + 3s^2 + 9)}{s^2(7s^2 + 6)}$
 $P = 0, \pm \sqrt{3}j, \pm 1.73j$

iii) $F(s) = k s \frac{(s^2+4)}{(s^2+1)(s^2+3)}$ $Z = 0, \pm 2j$
 $P = \pm j, \pm \sqrt{3}j$

LC N/w Synthesis Rules

- i) Foster 1st & 2nd form
- ii) Cauer 1st & 2nd form

Foster Form

- * 1st form is valid for $Z(s)$
 - * 2nd form is valid for $Y(s)$
- standard ϵs^4

$$Z(s) = \frac{k_0}{s} + \sum_{i=1}^n \frac{2k_i s}{s^2 + \omega_i^2} + HS$$

let $z(s)$ given

$$Z(s) = \frac{k(s^2+1)(s^2+4)}{s(s^2+4)} \text{ where } k=2$$

$$\frac{k_0}{s} \checkmark$$

check Pole at ∞ $N > D$
 Pole at 0
 Pole Complex

$$\frac{2k_i s}{s^2 + \omega_i^2} \checkmark$$

$$HS \checkmark$$

$$Z(s) = \frac{2(s^2+1)(s^2+4)}{s(s^2+4)} = \frac{k_0}{s} + \frac{2k_1 s}{s^2+4} + HS$$

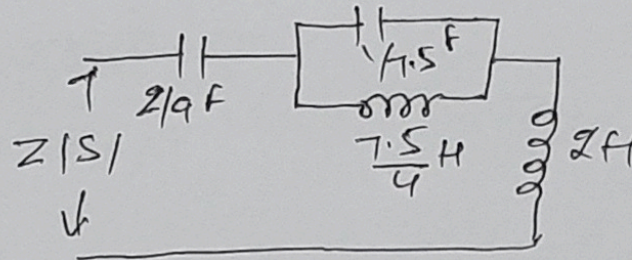
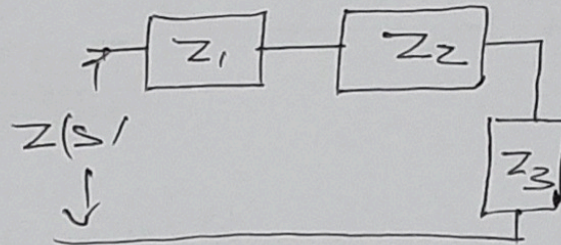
$$2(s^2+1)(s^2+4) = k_0(s^2+4) + 2k_1 s^2 + HS^2(s^2+4)$$

$$2s^4 + 20s^2 + 18 = HS^4 + (k_0 + 2k_1 + 4H)s^2 + 4k_0$$

$$H=2, \quad 4k_0=18 \quad k_0 + 2k_1 + 4H = 20$$

$$k_0 = 9/2, \quad k_1 = 15/4$$

$$Z(s) = \underbrace{\frac{9/2}{s}}_{Z_1} + \underbrace{\frac{15/2 s}{s^2+4}}_{Z_2} + \underbrace{2s}_{Z_3}$$



$$Z_3 = 2s \quad \left| \quad \frac{1}{Cs} \right|$$

$$Z_1 = \frac{9/2}{s} \quad \left| \quad \frac{1}{Cs} \right|$$

$$= \frac{1}{\frac{2}{9}s} \quad \left| \quad \frac{1}{Cs} \right|$$

$$C = \frac{2}{9} \text{ f}$$

$$Z_2 = \frac{15/2 s}{s^2+4} \quad \left| \quad \frac{Cs}{1/s} \right|$$

$$Y_2 = \frac{s^2+4}{7.5s} \quad \left| \quad \frac{Cs}{1/s} \right|$$

$$= \frac{s}{7.5} + \frac{4}{7.5s}$$

$$= \frac{1}{7.5} s + \frac{1}{7.5 \cdot \frac{4}{s}}$$

$$\frac{1}{7.5} \quad \frac{1}{1.5/4} H$$

Foster 2nd for LC N/w

$$Z(s) = \frac{s(s^2+1)(s^2+9)}{s(s^2+4)}$$

It is valid for $Y(s)$

$$Y(s) = \frac{s(s^2+4)}{s(s^2+1)(s^2+9)}$$

$$P=0 \quad X \rightarrow \frac{K_0}{s} \quad X$$

$$P=\infty \quad X \rightarrow \frac{H}{s} \quad X$$

$$P = \text{imaginary} \rightarrow \frac{2K_1 s}{s^2+1}, \frac{2K_2 s}{s^2+9}$$

$$Z(s) = Y(s) = \frac{s(s^2+4)}{s(s^2+1)(s^2+9)} = \frac{2K_1 s}{s^2+1} + \frac{2K_2 s}{s^2+9}$$

$$= 0.5 s(s^2+4) = 2K_1 s(s^2+9) + 2K_2 s(s^2+1)$$

$$0.5 s^3 + 2s = 2K_1 s^3 + 18K_1 s + 2K_2 s^3 + 2K_2 s$$

$$0.5 s^3 + 2s = (2K_1 + 2K_2) s^3 + (18K_1 + 2K_2) s$$

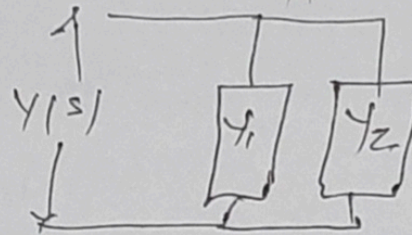
$$2K_1 + 2K_2 = 0.5$$

$$18K_1 + 2K_2 = 2$$

$$K_1 = 0.093$$

$$K_2 = 0.156$$

$$Y(s) = \frac{0.186 s}{s^2+1} + \frac{0.312 s}{s^2+9}$$



$$Y_1 = \frac{0.186 s}{s^2+1}$$

$$Z_1 = \frac{s^2+1}{0.186 s}$$

$$\left| \begin{array}{l} L s \\ \frac{1}{C s} \end{array} \right.$$

$$= \frac{s}{0.186 s} + \frac{1}{0.186 s}$$

$$= \frac{s}{0.186} + \frac{1}{0.186 s}$$

$$\frac{1}{s} \quad 5.37 H$$

$$\frac{1}{s} \quad 0.186 F$$

$$Y_2 = \frac{0.312 s}{s^2+9}$$

$$Z_2 = \frac{s^2+9}{0.312 s}$$

$$\left| \begin{array}{l} L s \\ \frac{1}{C s} \end{array} \right.$$

$$= \frac{s^2}{0.312 s} + \frac{9}{0.312 s}$$

$$= \frac{s}{0.312} + \frac{1}{0.312 \frac{9}{s}}$$

$$\frac{1}{s} \quad 0.312 H$$

$$\frac{1}{s} \quad 0.312 F$$

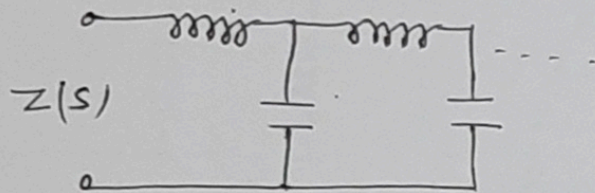
Cover 1st form of LC N/w

- * Use continued fraction (Power ↓ order)
- * Series element is always inductor
- * Parallel " " " Capacitor

$$Z(s) = Ls, \quad \frac{1}{Cs}$$

$$Y(s) = Cs, \quad \frac{1}{Ls}$$

General ckt for cover 1st form

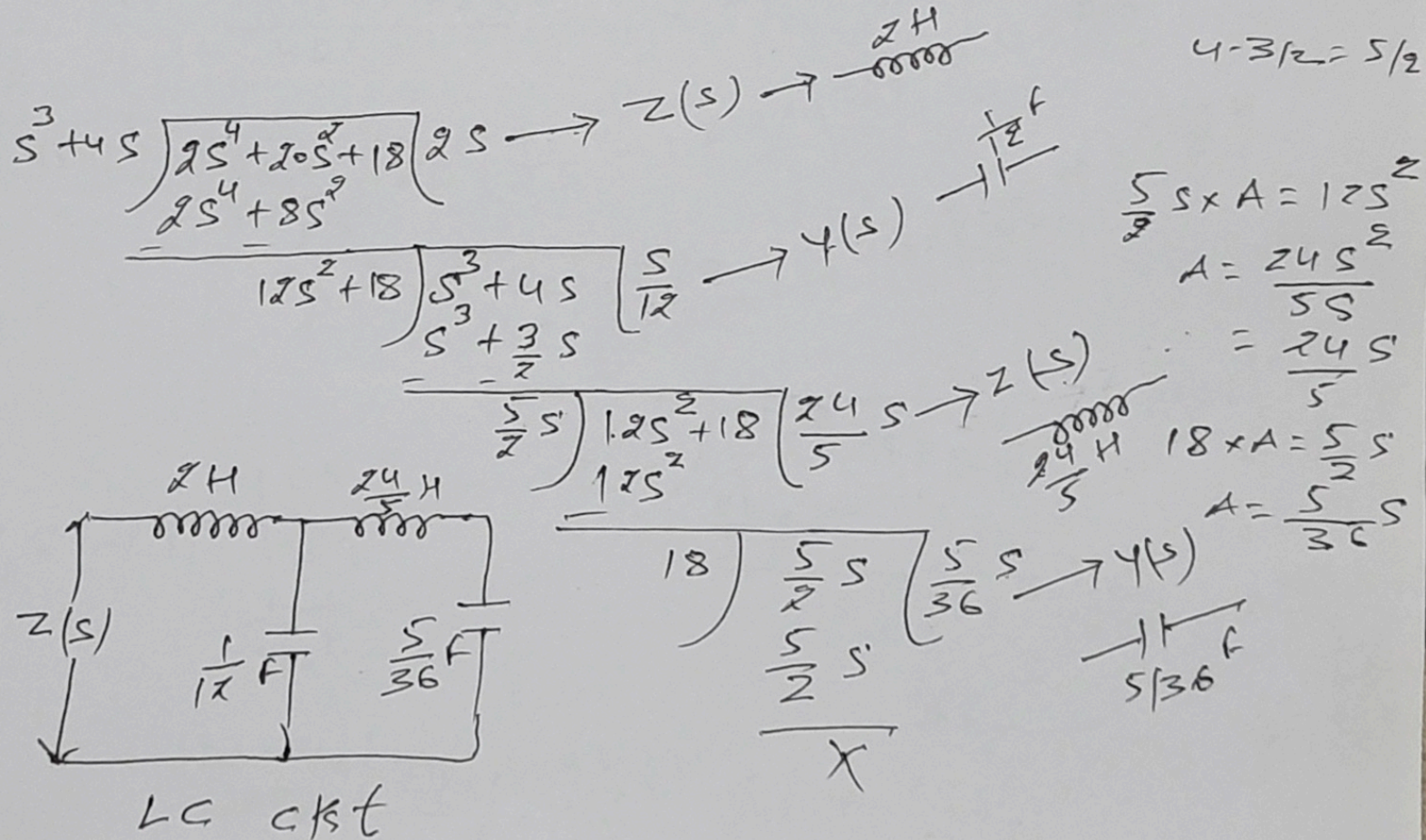


Example

$$Z(s) = \frac{2(s^2+1)(s^2+9)}{s(s^2+4)}$$

$$Z(s) = \frac{2(s^4+10s^2+9)}{s^3+4s}$$

$$Z(s) = \frac{2s^4+20s^2+18}{s^3+4s}$$



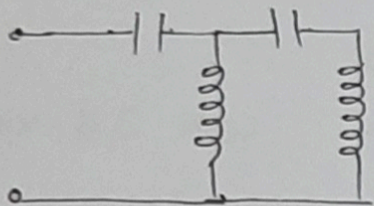
Cover 2nd form of LC H/W

* Use Continued fraction (Use Power in ↑ order)

* Series element → $\text{---} \text{---} \text{---}$

* Parallel element → $\text{---} \text{---} \text{---}$

General circuit



Example

$$Z(s) = \frac{s(s^2 + 4)}{2s^4 + 20s^2 + 18}$$

$$Y(s) = \frac{4s + s^3}{18 + 20s^2 + 2s^4}$$

$$18 + 20s^2 + 2s^4 \overline{) 4s + s^3} \begin{array}{r} \frac{4}{18} s \\ \underline{4s + \frac{80}{18}s^3 + \frac{8s^5}{18}} \\ -ve \end{array}$$

$$Z(s) = \frac{2s^4 + 20s^2 + 18}{s^3 + 4s}$$

$$Z(s) = \frac{18 + 20s^2 + 2s^4}{4s + s^3}$$

$$4s + s^3$$

$$\frac{18 + 20s^2 + 2s^4}{18 + \frac{9}{2}s^2} \left(\frac{9}{2s} \right) \rightarrow Z(s) \Big| \frac{1}{Cs} \Rightarrow C = \frac{2}{9} F$$

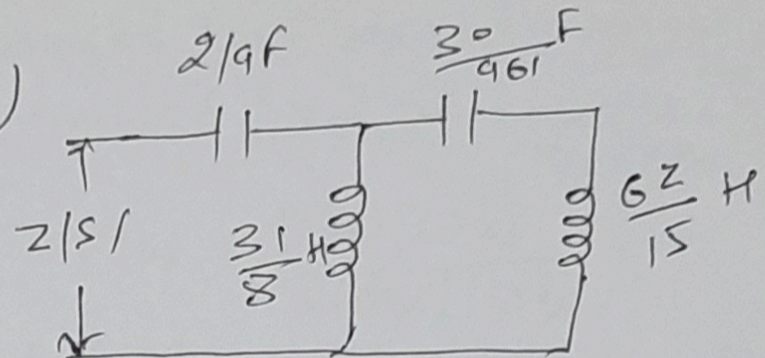
$$\frac{\frac{31}{2}s^2 + 2s^4}{4s + \frac{16}{31}s^3} \left(\frac{8}{31s} \right) \rightarrow Y(s) \Big| \frac{1}{Ls} \Rightarrow L = \frac{31}{8} H$$

$$\frac{\frac{15}{31}s^3}{\frac{31}{2}s^2 + 2s^4} \left(\frac{961}{30s} \right) \rightarrow Z(s) \Big| \frac{1}{Cs} \Rightarrow C = \frac{30}{961} F$$

$$\frac{2s^4}{\frac{15}{31}s^3 + \frac{15}{31}s^3} \left(\frac{15}{62s} \right) \rightarrow Y(s) \Big| \frac{1}{Ls} \Rightarrow L = \frac{62}{15} H$$

$$\frac{20 - 9/2}{3/2} \Big| \frac{31s^4}{2} \times A = 4s \quad A = \frac{4s \times 2}{31s^2}$$

$$\frac{15}{31}s^3 \times A = \frac{31}{2}s^2 \quad A = \frac{961}{30s^5}$$



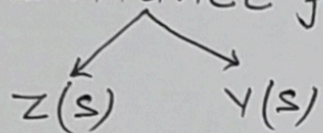
LC ckt ✓

RC or RL Network Synthesis

* RC Network:- R, C but not L (~~inductor~~)

* RL Network:- R, L but not C (~~capacitor~~)

* Given:- Immittance function



for RC N/w:- Z_{RC}, Y_{RC}

for RL N/w:- Z_{RL}, Y_{RL}

* $Z_{RC}(s)$ & $Y_{RL}(s)$ have same Properties

* $Y_{RC}(s)$ & $Z_{RL}(s)$ have same Properties

Properties of Z_{RC} & Y_{RL} N/w function.

- i) Poles & zeros lie on -ve real axis.
- ii) Poles & zeros must be alternate along the negative Real axis
- iii) The Residues of Z_{RC} & Y_{RL} must be real and Positive.

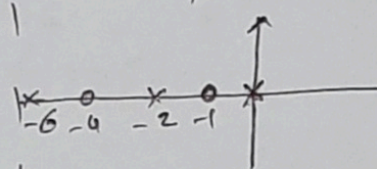
iv) lowest freq near to origin is Pole

v) Highest freq must be zero $Z_{RC}, Y_{RL} \xrightarrow{s=0} \infty$
 $Y_{RL} \text{ or } Z_{RC} \xrightarrow{s \rightarrow \infty} 0$

(e.g):- $F(s) = \frac{(s+1)(s+4)}{s(s+2)(s+6)}$

$P = 0, -2, -6$

$Z = -1, -4$



$$\frac{(s+1)(s+4)}{s(s+2)(s+6)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+6}$$

$$(s+1)(s+4) = A(s+2)(s+6) + Bs(s+6) + Cs(s+2)$$

$$s^2 + 5s + 4 = A(s^2 + 8s + 12) + Bs^2 + 6Bs + Cs^2 + 2Cs$$

compare Co-eff

$$s^2 + 5s + 4 = (A+B+C)s^2 + (8A+6B+2C)s + 12A$$

$$4 = 12A$$

$$A = 1/3$$

$$A+B+C = 1$$

$$8A+6B+2C = 5$$

Real, +ve

$$F(0) = \frac{1 \times 4}{0(2)(6)}$$

$$= \frac{4}{0} = \infty$$

$$B+C = 2/3$$

$$6B+2C = 7/3$$

$$B = 1/4$$

$$C = 0.11$$

$$A = 1/3$$

$$F(\infty) = \frac{s^2(1+\frac{1}{s})(1+\frac{4}{s})}{s^3(1+\frac{2}{s})(1+\frac{6}{s})}$$

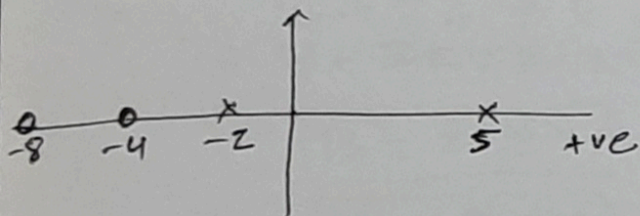
$$= \frac{1}{\infty} = 0$$

Properties of Z_{RL} & Y_{RC} N/w function

- i) Poles & zeros lie on negative Real axis.
- ii) Poles & zeros must be in alternate along -ve Real axis.
- iii) The Residues of Z_{RL} & Y_{RC} are Real, -ve.
then take $\frac{Z_{RL}}{s}$, $\frac{Y_{RC}}{s}$ now Residues will be Real & +ve.
- iv) Nearest to origin must be zero
It means Z_{RL} or $Y_{RC} \rightarrow 0$ with $s=0$
- v) Z_{RL} or $Y_{RC} \rightarrow \infty$ with $s=\infty$

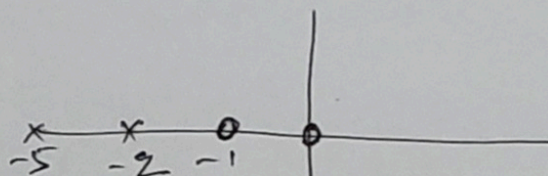
E.g. $\rightarrow$ ① $F(s) = \frac{(s+4)(s+8)}{(s+2)(s-5)}$

$Z = -4, -8$ X
 $P = -2, 5$



② $F(s) = \frac{s(s+1)}{(s+2)(s+5)}$

$Z = 0, -1$ X
 $P = -2, -5$



not in alternate

③ $F(s) = \frac{s(s+2)(s+6)}{(s+1)(s+4)}$ yes
 $Z = 0, -2, -6$ Z_{RL}
 $P = -1, -4$ Y_{RC}

$f(s) = \frac{s^3 + 8s^2 + 12s}{s^2 + 5s + 4}$

$\frac{s^3 + 8s^2 + 12s}{s^2 + 5s + 4} \left(\frac{s}{s^3 + 5s^2 + 4s} \right)$

$\frac{15}{8}$

$8 \overline{) 15} \begin{array}{r} 1 \\ 8 \\ \hline 7 \end{array}$

$1 + \frac{7}{8}$

$F(s) = s + \frac{3s^2 + 8s}{(s+1)(s+4)}$
 $g(s) = \frac{3s^2 + 8s}{(s+1)(s+4)}$

$g(s) = \frac{3s^2 + 8s}{(s+1)(s+4)}$

$\frac{s^2 + 5s + 4}{s^2 + 5s + 4} \left(\frac{3}{3s^2 + 15s + 12} \right)$

$g(s) = 3 + \frac{(-7s - 12)}{(s^2 + 5s + 4)}$

Residues Real, -ve

Foster 1 form of RC N/W

$$Z(s) = \frac{(s+1)(s+4)}{s(s+2)(s+5)} = Z_{RC}(s)$$

$$\frac{(s+1)(s+4)}{s(s+2)(s+5)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+5}$$

$$(s+1)(s+4) = A(s+2)(s+5) + Bs(s+5) + Cs(s+2)$$

$$s^2 + 5s + 4 = A(s^2 + 7s + 10) + Bs^2 + 5Bs + Cs^2 + 2Cs$$

compare Co-eff

$$s^2 + 5s + 4 = (A+B+C)s^2 + (7A+5B+2C)s + 10A$$

$$4 = 10A$$

$$A = 0.4$$

$$A+B+C = 1$$

$$B+C = 0.6 \quad \text{--- (I)}$$

$$7A+5B+2C = 5$$

$$5B+2C = 2.2 \quad \text{--- (II)}$$

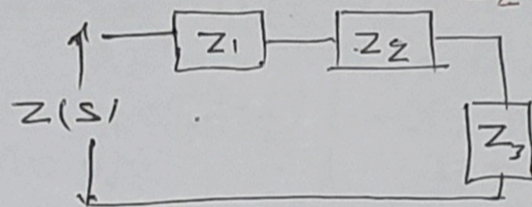
$$B+C = 0.6$$

$$5B+2C = 2.2$$

$$B = 0.33$$

$$C = 2.67 = 0.267$$

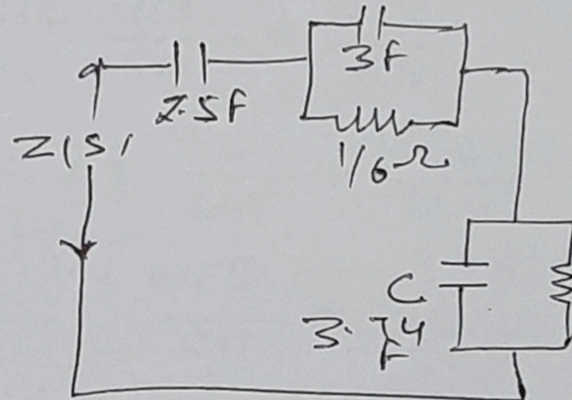
$$Z(s) = \frac{0.4}{s} + \frac{0.33}{s+2} + \frac{0.267}{s+5}$$



$$Z_1 = \frac{0.4}{s} \quad \left| \frac{1}{Cs} \right. \\ C = \frac{1}{0.4} = 2.5 F$$

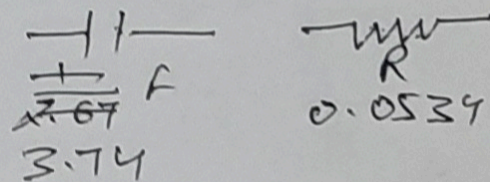
$$Z_2 = \frac{0.33}{s+2}$$

$$Y_2 = \frac{s+2}{0.33} \quad \left| Cs \right. \\ \frac{s}{0.33} + \frac{2}{0.33}$$



$$Z_3 = \frac{0.267}{s+5}$$

$$Y_3 = \frac{s+5}{0.267} \quad \left| Cs \right. \\ = \frac{s}{0.267} + \frac{5}{0.267}$$



Foster 2 form of RC N/W

$$Z(s) = \frac{(s+1)(s+4)}{s(s+2)(s+5)}$$

It is valid for $y(s)$

$$y(s) = \frac{s(s+2)(s+5)}{(s+1)(s+4)}$$

Y_{RC}
Residue -ve
Real.

$$\frac{y(s)}{s} = \frac{(s+2)(s+5)}{(s+1)(s+4)}$$

$$\frac{y(s)}{s} = \frac{s^2 + 7s + 10}{s^2 + 5s + 4}$$

$$\frac{s^2 + 5s + 4}{s^2 + 5s + 4} \left[\frac{s^2 + 7s + 10}{s^2 + 5s + 4} \right] = 2s + 6$$

$$\frac{15}{7} = 2 + \frac{1}{7}$$

$$\frac{y(s)}{s} = 1 + \frac{2s+6}{s^2+5s+4}$$

$$G(s) = \frac{2s+6}{(s+1)(s+4)}$$

$$G(s) = \frac{2s+6}{(s+1)(s+4)} = \frac{A}{s+1} + \frac{B}{s+4}$$

$$2s+6 = A(s+4) + B(s+1)$$

$$s = -1$$

$$4 = A(3)$$

$$A = 4/3$$

$$s = -4$$

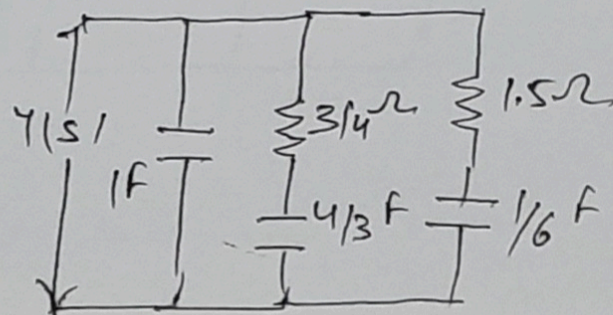
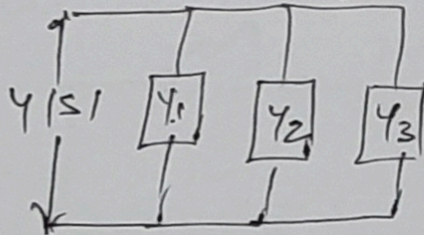
$$-2 = B(-3)$$

$$B = 2/3$$

$$G(s) = \frac{4/3}{s+1} + \frac{2/3}{s+4}$$

$$\frac{y(s)}{s} = 1 + \frac{4/3}{s+1} + \frac{2/3}{s+4}$$

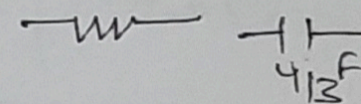
$$y(s) = s + \frac{4/3 s}{s+1} + \frac{2/3 s}{s+4}$$



$$y_2 = \frac{4/3 s}{s+1}$$

$$Z_2 = \frac{s+1}{\frac{4}{3} s}$$

$$= \frac{3}{4} + \frac{1}{\frac{4}{3} s}$$

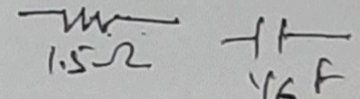


$$y_3 = \frac{2/3 s}{s+4}$$

$$Z_3 = \frac{s+4}{\frac{2}{3} s}$$

$$= \frac{3}{2} + \frac{4}{\frac{2}{3} s}$$

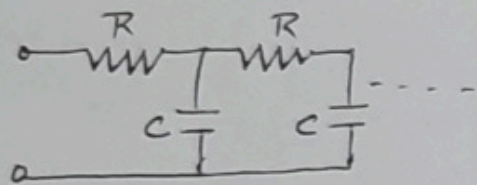
$$3/2 + \frac{1}{\frac{2}{12} s}$$



Cover 1 form of RC N/w

- * Power in descending order
- * Series element Resistor
- * Parallel element Capacitor

General circuit



$$s^2 + 5s + 4 \left) \frac{s^3 + 7s^2 + 10s}{s^3 + 5s^2 + 4s} \right. s \rightarrow y(s) \quad \frac{1f}{1} \text{ Parallel}$$

$$\frac{2s^2 + 6s}{s^2 + 3s} \left) \frac{s^2 + 5s + 4}{s^2 + 3s} \right. \left(\frac{1}{2} \rightarrow z(s) \right) \quad \frac{1/2 \Omega}{1} \text{ Series}$$

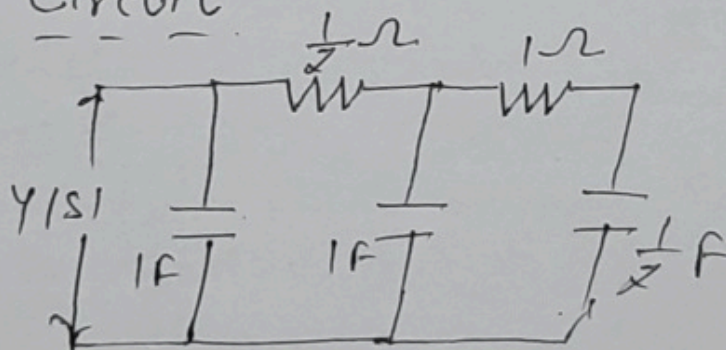
Eg. →

$$z(s) = \frac{(s+1)(s+4)}{s(s+2)(s+5)} = z_{RC}(s)$$

$$z(s) = \frac{s^2 + 5s + 4}{s^3 + 7s^2 + 10s}$$

$$y(s) = \frac{s^3 + 7s^2 + 10s}{s^2 + 5s + 4}$$

Circuit



$$\frac{2s+4}{2s^2+4s} \left) \frac{2s^2+6s}{2s^2+4s} \right. s \rightarrow y(s) \quad \frac{1f}{1}$$

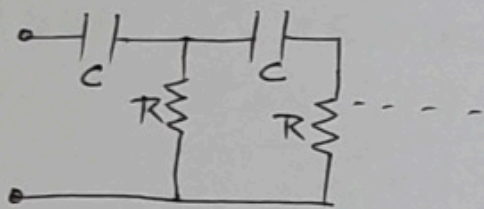
$$\frac{2s}{2s} \left) \frac{2s+4}{2s} \right. 1 \rightarrow z(s) \quad \frac{1/2 \Omega}{1}$$

$$\frac{4}{4} \left) \frac{2s}{2s} \right. \left(\frac{s}{2} \rightarrow y(s) \right) \quad \frac{1f}{1/2 f}$$

Cover & form of RC N/w

- * Power in ascending order
- * Series element: Capacitor
- * Parallel element: Resistor

General circuit



$$\epsilon g \rightarrow Z(s) = \frac{(s+1)(s+4)}{s(s+2)(s+5)} = Z_{RC}$$

$$Z(s) = \frac{s^2 + 5s + 4}{s^3 + 7s^2 + 10s}$$

$$Z(s) = \frac{4 + 5s + s^2}{10s + 7s^2 + s^3}$$

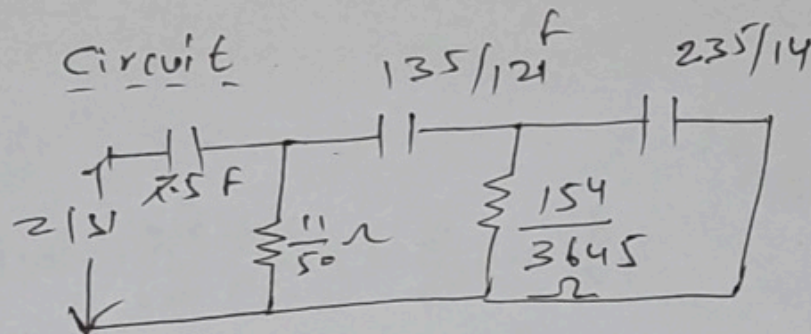
$$7 - \frac{30}{11}$$

$$\frac{3}{5} - \frac{121}{235}$$

$$\frac{141 - 121}{235}$$

47

Circuit



$$10s + 7s^2 + s^3 \sqrt{\frac{4 + 5s + s^2}{4 + \frac{14}{5}s + \frac{2}{5}s^2}} \left(\frac{4}{10s} \right) \rightarrow Z(s) \quad \frac{1}{Cs} = \frac{10}{4} = 25F \text{ series}$$

$$\frac{\frac{11}{5}s + \frac{3}{5}s^2}{10s + 7s^2 + s^3} \left(\frac{5}{11} \times 10 \right) \rightarrow Y(s) \quad \frac{11}{50} \Omega$$

$$\frac{\frac{47}{11}s^2 + s^3}{\frac{11}{5}s + \frac{3}{5}s^2} \left(\frac{11}{47} \times \frac{11}{5s} \right) \rightarrow Z(s) \quad C = \frac{235}{121}$$

$$\frac{\frac{20}{235}s^2}{\frac{47}{11}s^2} \left(\frac{235}{20} \times \frac{47}{11} \right) \rightarrow Y(s)$$

$$\frac{\frac{20}{235}s^2}{s^3} \left(\frac{20}{235s} \right) \rightarrow Z(s) \quad C = \frac{235}{20}$$

Foster 1 form of RL N/w

$$Z(s) = \frac{s(s+2)(s+5)}{(s+1)(s+4)}$$

Z_{RL} given

* Residues Real, -ve

$$\begin{array}{r} s^2 + 5s + 4 \overline{) s^3 + 7s^2 + 10s} \quad (s+2) \\ \underline{s^3 + 5s^2 + 4s} \\ 2s^2 + 10s + 8 \\ \underline{2s^2 + 10s + 8} \\ -4s - 8 \end{array}$$

-ve

$$\frac{Z(s)}{s} = \frac{(s+2)(s+5)}{(s+1)(s+4)}$$

$$\frac{Z(s)}{s} = \frac{s^2 + 7s + 10}{s^2 + 5s + 4} = 1 + \left[\frac{2s + 6}{(s+1)(s+4)} \right]$$

$\downarrow$
 $A(s)$

$$\begin{array}{r} s^2 + 5s + 4 \overline{) s^2 + 7s + 10} \quad (1) \\ \underline{s^2 + 5s + 4} \\ 2s + 6 \end{array}$$

$$\frac{15}{2} = 7 + \frac{1}{2}$$

$$2 \overline{) 15} \begin{array}{r} 7 \\ 14 \\ \hline 1 \end{array}$$

Z_{RL} or Y_{RC}

$$A(s) = \frac{2s + 6}{(s+1)(s+4)} = \frac{a}{s+1} + \frac{b}{s+4}$$

$$2s + 6 = a(s+4) + b(s+1)$$

$$s = -1, \quad 4 = a(3)$$

$$a = 4/3$$

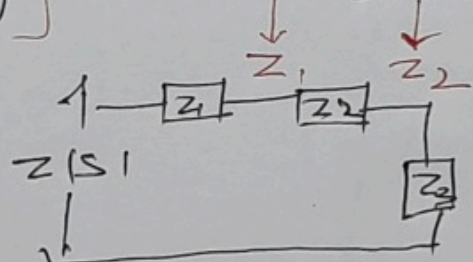
$$s = -4, \quad -2 = b(-3)$$

$$b = 2/3$$

$$A(s) = \frac{4}{3s+3} + \frac{2}{3s+12}$$

$$\frac{Z(s)}{s} = 1 + \frac{4}{3s+3} + \frac{2}{3s+12}$$

$$Z(s) = s + \frac{4s}{3s+3} + \frac{2s}{3s+12}$$



$$Z_1 = s \rightarrow \text{inductor}$$

$$Z_2 = \frac{4s}{3s+3}$$

$$Y_2 = \frac{3s+3}{4s} = \frac{3}{4} + \frac{3}{4s} = \frac{3}{4} + \frac{1}{\frac{4}{3}s}$$

$$Z_3 = \frac{2s}{3s+12}$$

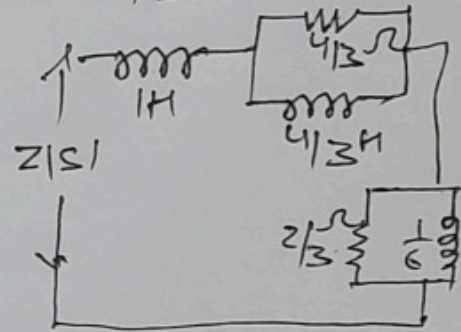
$$Y_3 = \frac{3s+12}{2s}$$

$$= \frac{3}{2} + \frac{12}{2s}$$

$$= \frac{3}{2} + \frac{1}{\frac{2}{12}s}$$

$$\uparrow$$

$$3/2 \Omega \quad 1/6 H$$



✓

Foster 2 form of RL N/W

$$Z(s) = \frac{s(s+2)(s+5)}{(s+1)(s+4)} = Z_{RL}(s)$$

Z_{RC} or Y_{RL}
Residue

F-2 form valid for $Y(s)$ function

$$Y(s) = \frac{(s+1)(s+4)}{s(s+2)(s+5)}$$

$$\frac{(s+1)(s+4)}{s(s+2)(s+5)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s+5}$$

$$(s+1)(s+4) = A(s+2)(s+5) + Bs(s+5) + Cs(s+2)$$

$$s=0 \rightarrow 4 = A(10) \Rightarrow A = 0.4$$

$$s=-2 \rightarrow -2 = B(-6) \Rightarrow B = 1/3$$

$$s=-5 \rightarrow -4 \times (-1) = C(-5)(-3) \Rightarrow C = 4/15$$

$$Y(s) = \frac{0.4}{s} + \frac{1/3}{s+2} + \frac{4/15}{s+5}$$

$\downarrow$ $\downarrow$ $\downarrow$
 Y_1 Y_2 Y_3

$$Y_1 = \frac{0.4}{s} = \frac{1}{2.5s} \Rightarrow L = \frac{10}{4} = 2.5 H$$

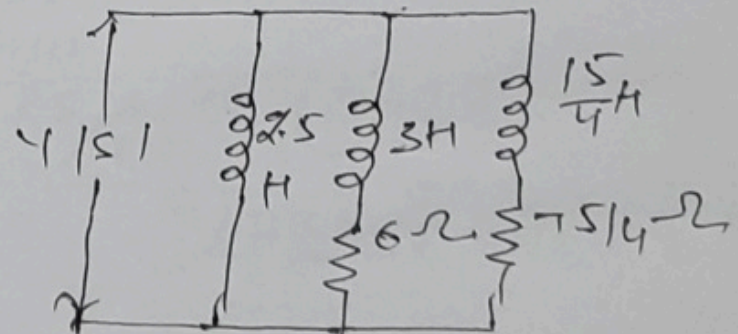
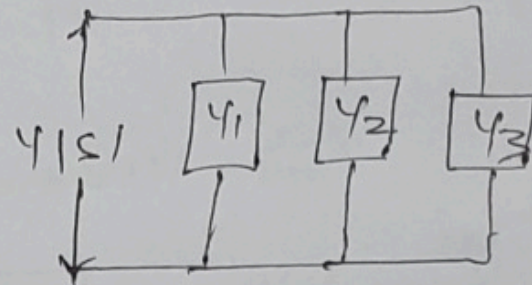
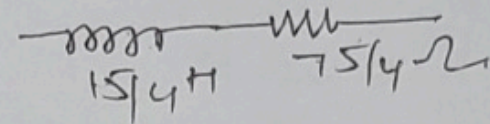
$$Y_2 = \frac{1}{3s+6}$$

$$Z_2 = 3s+6 \Rightarrow \frac{3H}{6\Omega}$$

$$Y_3 = \frac{4}{15s+75}$$

$$Z_3 = \frac{15s+75}{4} = 3.75s + 18.75$$

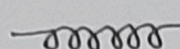
$= Ls + R$



✓

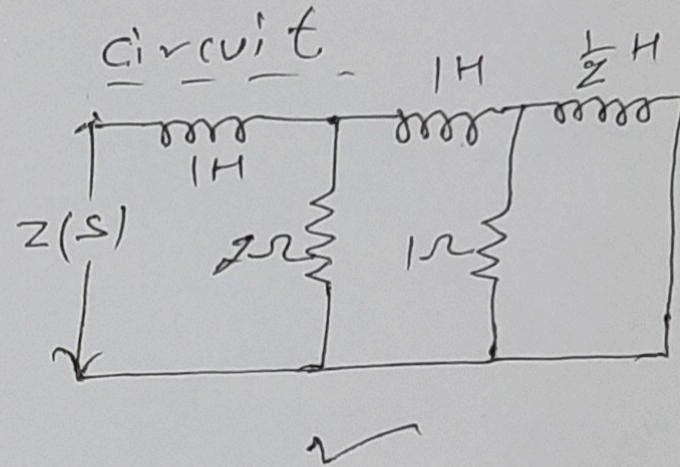
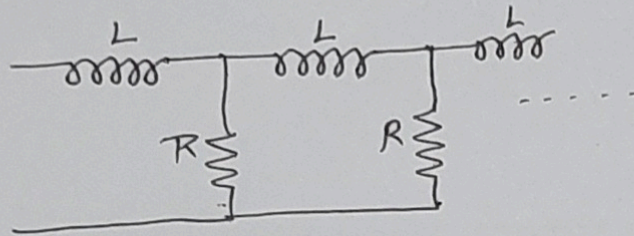
Cover 1st form of RL N/w

* Use Polynomial in descending order

* Series element of Ladder N/w $\rightarrow$  (L)

* Parallel " " " " $\rightarrow$  (R)

General circuit



example

$$Z(s) = \frac{s(s+2)(s+5)}{(s+1)(s+4)}$$

$$Z(s) = \frac{s^3 + 7s^2 + 10s}{s^2 + 5s + 4}$$

$$\begin{array}{r} s^2 + 5s + 4 \overline{) s^3 + 7s^2 + 10s} \\ \underline{s^3 + 5s^2 + 4s} \\ 2s^2 + 6s \end{array} \quad s \rightarrow Z(s) = 1H$$

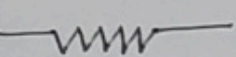
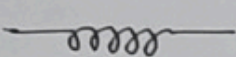
$$\begin{array}{r} 2s^2 + 6s \overline{) 2s^2 + 6s} \\ \underline{2s^2 + 6s} \\ 0 \end{array} \quad \frac{1}{2} \rightarrow 4\Omega$$

$$\begin{array}{r} 2s^2 + 6s \overline{) 2s^2 + 6s} \\ \underline{2s^2 + 6s} \\ 0 \end{array} \quad s \rightarrow Z(s) = 1H$$

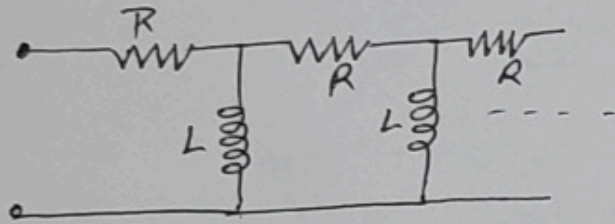
$$\begin{array}{r} 2s \overline{) 2s} \\ \underline{2s} \\ 0 \end{array} \quad 1 \rightarrow 1\Omega$$

$$\begin{array}{r} 4 \overline{) 2s} \\ \underline{2s} \\ 0 \end{array} \quad \frac{s}{2} \rightarrow \frac{1}{2}H$$

Cover 2nd form of RL N/w

- * Use Polynomial in Ascending order
- * Series element $\rightarrow$  R
- * Parallel element $\rightarrow$  L

General circuit



Example

$$Z(s) = \frac{s(s+2)(s+5)}{(s+1)(s+4)}$$

$$Z(s) = \frac{s^3 + 7s^2 + 10s}{s^2 + 5s + 4}$$

$$Z(s) = \frac{10s + 7s^2 + s^3}{4 + 5s + s^2}$$

$$4 + 5s + s^2 \overline{) \begin{array}{r} 10s + 7s^2 + s^3 \\ 10s + 12.5s^2 + 2.5s^3 \\ \hline -5.5s^2 - 1.5s^3 \end{array}} \frac{10}{4} s$$

-ve



$$Y(s) = \frac{4 + 5s + s^2}{10s + 7s^2 + s^3}$$

$$10s + 7s^2 + s^3 \overline{) 4 + 5s + s^2} \left(\frac{4}{10s} \rightarrow Y(s) = \frac{1}{Ls} \Rightarrow L = \frac{10}{4} \text{ H} \right)$$

$$2.2s + 0.6s^2 \overline{) 10s + 7s^2 + s^3} \left(\frac{10}{2.2} \rightarrow Z(s) = \frac{10}{2.2} \Omega \right)$$

$$10s + 2.7s^2 + \dots \overline{) 10s + 7s^2 + s^3} \left(\frac{10}{2.2} \rightarrow Z(s) = \frac{10}{2.2} \Omega \right)$$

$$4.27s^2 + s^3 \overline{) 2.2s + 0.6s^2} \left(\frac{2.2}{4.27s} \rightarrow Y(s) \right)$$

$$0.09s^2 \overline{) 4.27s^2 + s^3} \left(\frac{4.27}{0.09} \rightarrow Z(s) \right)$$

$$s^3 \overline{) 0.09s^2} \left(\frac{0.09/s}{0.09s^2} \rightarrow Y(s) \right)$$

$$L = \frac{1}{0.09} \text{ H}$$

